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- e. Verify nameplate/type labelled and system ordered, include check on input/output voltages and rating.
- f. Confirm cabinet/UPS and accessories (battery/racks) dimensions as design layout/footprint. Ensure compliance of minimum free access, maintenance and ventilation free space meets the local regulation, specification and manufacturer requirement.
- g. Connect the UPS and check voltages and phase rotation (for 3 phase unit) of mains input voltage.
- h. Check earthing/grounding system and earth/ground connection before powering up the UPS.
- i. Connect all the battery cables, output/load cables, alarm cables, etc., before powering up UPS and commissioning it in accordance with drawings/instruction manual provided.

6.7 Transformers

6.7.1 General requirement

- a) The heating and cooling systems shall be sized to ensure that while operating during normal conditions, the environment in the space or room for the transformer will be maintained within the transformer manufacturer's recommended ambient temperature specifications.
- b) Adequate space shall be provided around the transformer to allow for air circulation and natural ventilation.
- c) Transformers shall be protected against potential condensation, leaks, spills, or other liquids.
- d) Fire suppression systems shall meet all applicable local codes and standards.
- e) Equipment shall be braced to meet seismic requirements and regulations.
- f) Transformers serving non-sinusoidal loads shall be designed with adequate K factor ratings.
- g) The enclosure and fittings shall provide adequate protection from the environment where it is installed.
- h) Transformer design shall meet seismic requirements for the locale where it is being installed.
- i) Transformers shall be suitable for switching by all types of switching devices, i.e., vacuum or gas breakers without detrimental effect on the windings.
- j) The manufacturer shall be responsible to ensure the unit has been fully derated to function in the site conditions.
- k) Structure loading shall be reviewed before selection and placement of transformers.

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6.7.2 Gas or Liquid-filled type HV main transformer

6.7.2.1 Specific requirement

- i. kVA ratings of transformers specified in data sheets are continuous and based on not exceeding the temperature rise limits.
- ii. kVA rating is based on rated secondary voltage, being maintained by impressing on the primary tap indicated a voltage higher and its rated voltage to compensate the effect of regulation of the transformer.
- iii. The transformer shall deliver rated kVA continuously at 105% voltage and operate continuously at 10% above rated voltage at no-load, without exceeding the specified temperature rises.
- iv. When force-cooled ratings are specified, all current carrying parts including bushings shall be sized for highest rating.
- v. Oil-immersed power transformer shall use FR3 fluid type for its Tank and OLTC Tank.
- vi. Average temperature rise by resistance test for the oil-immersed power transformer shall not exceed 65 deg C based on an average ambient of 30 deg C for any 24-hour period.
- vii. The transformer impedance is measured at self-cooled rating and shall be specified in data sheet.
- viii. Core Iron loss at fundamental frequency 50/60Hz shall be preferably less than 1.25 watts/kg.
- ix. The level of noise produced by the transformer, operating at rated voltage and frequency, shall be below limitations which required by NEMA TR-1.
- x. Transformer windings and core shall be braced to be able withstand without damage thereunto the electromagnetic stresses produced by the maximum short circuit current of 2 Second.
- xi. Insulation voltage class and related BIL-basic impulse insulation level for each transformer shall be specified in data sheet.
- xii. The core shall be built up of cold rolled, oriented grain non-aging silicon steel laminations having a high permeability and low hysteresis loss.
- xiii. The core shall be rigidly clamped and braced to resist distortion caused by short circuit stresses or transportation handling, and to prevent the shifting of core laminations with the resulting

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deterioration or damage to winding insulation and laminations.

- xiv. Winding conductors shall be Cooper material. It shall be free from scale, burrs and splinters, properly rounded corners and uniformly insulated.
- xv. Turns shall be wound, and coils shall be formed and braced so that finished windings are rigid, capable of withstanding short circuit stresses without deflection injurious to the insulation, and free from the hazards of sagging or slipping of turns.

6.7.2.2 Ratings

- i. The application/duty of the transformer will be mentioned in the requisition. Example generator transformer, motor unit transformer, etc.,
- ii. Power transformer use for supply to group of motors shall be capable of withstanding in frequent restarting loads of up to 1.8 times the rated transformer current.
- iii. IEC60076-1 or IEEE C57-12 or equivalent standard rated transformer should be preferred.
- iv. Percentage Impedance voltage based on rated power should be as follows, unless an overriding design requirement.
 - Impedance voltages for transformers rating above 25,000kVA shall be specified in the requisition, depending on system design and/or system studies

Rated Power (kVA)	Impedance Voltage (%)
Up to 630	4.00
631 to 1,250	5.00
1,251 to 3,150	6.25
3,151 to 6,300	7.15
6,301 to 12,500	8.35
12,501 to 25,000	10.00

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v. Transformer and System Voltage

- o Rated voltage on the high voltage side is generally equal to the system nominal voltage in which the high voltage winding is connected.
- o The correlation between the rated voltage (at no load) of the low voltage winding and the corresponding system voltage is as below:

System nominal Voltage (V)	Rated Secondary Voltage (No Load)
200	208
208	214
380	400
400	420
415	433
440	460
480	504
3300	3450
6600	6900
11000	11500
33000	34500
66000	72500

vi. Nameplate Details:

Rating plate shall be stainless steel or others that is corrosion resistance metal. It should be permanently fixed to a readily accessible and non-removable part of the transformer, such as cooling fins.

Nameplate shall have the following information:

- o Name of Manufacturer
- o Serial number
- o Order No./Project number
- o Rated Capacity: kVA/MVA Rating
- o Primary voltage/s at various transformer tapping/connections
- o Secondary voltage

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- o Full load currents (Primary & Secondary)
- o Rated Frequency
- o Number of Phases
- o Vector group
(connection/displacement)/Diagram
- o Connection Diagram
- o Impedance rating/Percentage Impedance
- o Cooling of transformer (e.g. ONAN/ONAF)
- o Protection category/type (e.g. NEMA 1
Enclosure/IPxx)
- o Quantity of insulating gas/liquid (litres/gallons
of oil) and type
- o Insulation/temperature class of winding
insulation.
- o Manufacturer tested date
- o Total weight (kg)

6.7.2.3 On-load Tap Changer (OLTC)

- i. Main transformer shall be equipped with integrated on-load tap changer placed in the primary windings. Each tap of OLTC shall be continuously operated under +/- 2.5% variation of Primary incoming voltage.
- ii. Overall tapping range +/- 1.25% (+/- 19 steps)
- iii. Maximum step voltage 2.5%
- iv. Tolerance over tapping range +/- 0.5% of rated voltage
- v. The tap changer shall be installed in separate oil compartment in the transformer casing
- vi. The OLTC shall have local and remote control with local manual operating devices. When the manual operating device is in use, the remote control system shall not influence the tap changer.
- vii. The tap changer shall be capable of automatic or manual control. It shall be mounted on the transformer casing and at least consist of: electrical supply, electrical motor, drive shaft for tap changer, mechanical counter w/ reset button, auxiliary relays/contractor/limit switches, space heater w/ thermostat.

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6.7.2.4 Other main transformer parts and accessories

- i. Bushings
- ii. Terminal Chamber
- iii. Tank
- iv. Oil preservation system
- v. Insulation Oil
- vi. Forced cooling device
- vii. Pressure relieve device
- viii. Bucholz relay or Sudden pressure relay
- ix. Gas detector relay
- x. Gas sampling device
- xi. Common terminal box
- xii. Liquid level gauge
- xiii. Oil thermometer
- xiv. Pressure-vacuum gauge and bleeder
- xv. Globe type combination drain and upper/lower filter valve with sampling device
- xvi. Lifting eyes for cover
- xvii. Facilities for lifting core and coil assembly from tank
- xviii. Lugs for lifting, jacking facilities, moving facilities.
- xix. Tank grounding pads
- xx. Nameplates
- xxi. On-line oil purifier and filter equipment
- xxii. On-line Partial discharge measurement sensor and system (if required)
- xxiii. On-line dissolved gas analysis monitoring.

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6.7.2.5 Control panel (OLTC)

- i. Tap changer remote control panel shall be located indoor. Tap changer local panel outdoor use with enclosure protection IP55.
- ii. A remote-control panel for remote and automatic operation and indication of tap changer shall be supplied by qualified maker that proposed by Transformer manufacturer and approved by local authority company.
- iii. Control panel shall be equipped with important apparatus such as: remote-local change over switch, push button for tap position 'raise/lower', manual tap changing mechanism, mechanical tap position indicator, AVR relay, indication lamp, PTT, CTT, etc.
- iv. AVR-Automatic Voltage Regulating relay shall be intelligent electronic type with high adjustable capability for set point range, sensitivity, pick-up time, under voltage blocking and line drop compensator.

6.7.2.6 Control power

- i. Control power shall be supplied via DC power supply which specified in datasheet and details in section 3.6

6.7.3 Liquid-filled/Dry-type cast resin type MV distribution transformer

6.7.3.1 Specific requirement

- i. kVA ratings of transformers specified in data sheets are continuous and based on not exceeding the temperature rise limits.
- ii. kVA rating is based on rated secondary voltage, being maintained by impressing on the primary tap indicated a voltage higher and its rated voltage to compensate the effect of regulation of the transformer.
- iii. The transformer shall deliver rated kVA continuously at 105% voltage and operate continuously at 10% above rated voltage at no-load, without exceeding the specified temperature rises.
- iv. When forced-air cooled ratings are specified, all current carrying parts including bushings shall be sized for highest rating.

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- v. Oil-immersed distribution transformer shall use less flammable FR3 fluid.
- vi. Average temperature rise by resistance test for the oil-immersed distribution transformer shall not exceed 65oC based on an average ambient of 30oC for any 24-hour period.
- vii. Average temperature rise by resistance test for the dry-type distribution transformer shall not exceed 80oC based on an average ambient of 30oC for any 24-hour period.
- viii. The transformer impedance is measured at self-cooled rating and shall be specified in data sheet.
- ix. The level of noise produced by the transformer, operating at rated voltage and frequency, shall be below limitations which required by NEMA TR-1.
- x. Transformer windings and core shall be braced to be able withstand without damage thereunto the electromagnetic stresses produced by the maximum short circuit current of 2 Second.
- xi. Insulation voltage class and related BIL-basic impulse insulation level for each transformer shall be specified in data sheet.
- xii. The core shall be built up of cold rolled, oriented grain non-aging silicon steel laminations having a high permeability and low hysteresis loss.
- xiii. The core shall be rigidly clamped and braced to resist distortion caused by short circuit stresses or transportation handling, and to prevent the shifting of core laminations with the resulting deterioration or damage to winding insulation and laminations.
- xiv. Winding conductors shall be Cooper material. It shall be free from scale, burrs and splinters, properly rounded corners and uniformly insulated.
- xv. Turns shall be wound, and coils shall be formed and braced so that finished windings are rigid, capable of withstanding short circuit stresses without deflection injurious to the insulation, and free from the hazards of sagging or slipping of turns.

6.7.3.2 Ratings

(Refer to 6.7.2.2)

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6.7.3.3 Accessories

- i. Provide fan cooling equipment (fans, power/control wiring, Local selector switch, controller, fuse, indicating light, relay)
- ii. Three-phase winding temperature indicator with contacts to control fans and alarm contacts
- iii. Provide Class B Stranded Type SIS power/control conductors
- iv. Copper ground pads
- v. Power monitoring connection
- vi. Nameplates
- vii. Grounding resistor as required by designer and specified in data sheet
- viii. Bushing Current Transformer type as required by designer
- ix. For liquid filled distribution transformer, shall provide additional item as minimum.
 - o Pressure-vacuum gauge with bleeder
 - o Liquid-level gauge with alarm contacts
 - o Dial-type temperature gauge with alarm contacts
 - o Drain valve with sample port
 - o Fill plug
 - o Pressure relief device
 - o Sudden pressure relay
 - o Winding Temperature relay

6.7.3.4 Control power

- a) Self-powered from secondary bushing single phase

6.7.4 Dry-type cast resin type LV distribution transformer

6.7.4.1 Specific requirement

- i. Transformers shall be continuously rated for maximum output in all operating modes (ONAN).
- ii. The maximum design demand load allowance for redundant units shall be 40% and 80% for non-redundant units.
- iii. Transformers used for redundancy or maintenance activities shall be suitable for parallel operation.
- iv. The transformer core shall be constructed of high-grade, non-aging, silicon steel laminations with

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- high magnetic permeability and low hysteresis and eddy current losses.
- v. The transformer shall be constructed of individually cast primary and secondary coils, and coaxially mounted.
 - vi. The high voltage winding shall be vacuum cast in epoxy with the low voltage windings wound separately and be either pressure injected, vacuum cast, or hermetically sealed with foil or sheet conductors.
 - vii. Insulation shall be Class F and shall incorporate intrinsic fire extinguishing properties.
 - viii. Random wound coils are not acceptable for use in medium voltage windings.
 - ix. Windings shall have minimal internal splices or joints.
 - x. Transformers serving non-sinusoidal loads shall be designed with adequate K factor ratings, i.e. K-4 type for facility load and K-13 type for process load are required base on harmonic study to verify the suitable type of transformer.
 - xii. Transformers door (HV side and LV side) should not be equipped with interlock function with its upstream and downstream protection device (Circuit breaker), it includes transformers door opening/closing shall not interrupt normal operation. If required by local code or regulation, the transformers doors opening can provide with an alarming function (sent via SCADA or locally sounded) to inform the site the transformers door has been opened. It is further recommended any trip interlock function associated with transformers door access to be disabled, unless required by Local Code or Regulation.

6.7.4.2 Ratings

- i. The application/duty of the transformer will be mentioned in the requisition. Example motor unit transformer, transformer for variable speed drive etc.,
- ii. Power transformer use for supply to group of motors shall be capable of withstanding in frequent restarting loads of up to 1.8 times the rated transformer current.
- iii. IEC60076-1 or IEEE C57-12 or equivalent standard rated transformer should be preferred. In order to

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limiting the LV switchgear capacity to 5000A Facility or Production transformers maximum rating should be as follows:

MV/208V – 2MVA,

MV/480V – 4MVA.

- iv. Percentage Impedance voltage based on rated power should be as follows, unless an overriding design requirement.

Rated Power (kVA)	Impedance Voltage (%)
Up to 630	4.00
631 to 1,250	5.00
1,251 to 3,150	6.25
3,151 to 6,300	7.15
6,301 to 12,500	8.35
12,501 to 25,000	10.00

Impedance voltages for transformers rating above 25,000kVA shall be specified in the requisition, depending on system design and/or system studies

- v. Transformer and System Voltage
- o Rated voltage on the high voltage side is generally equal to the system nominal voltage in which the high voltage winding is connected.
 - o The correlation between the rated voltage (at no load) of the low voltage winding and the corresponding system voltage is as below:

System nominal Voltage (V)	Rated Secondary Voltage (No Load)
200	208
208	214
380	400
400	420
415	433
440	460
480	504

vi. Nameplate Details:

Rating plate shall be stainless steel or others that is corrosion resistance metal. It should be permanently fixed to a readily accessible and non-removable part of the transformer, such as cooling fins.

Nameplate shall have the following information:

- o Name of Manufacturer
- o Serial number
- o Order No./Project number
- o Rated Capacity: kVA/MVA Rating
- o Primary voltage/s at various transformer tapping/connections
- o Secondary voltage
- o Full load currents (Primary & Secondary)
- o Rated Frequency
- o Number of Phases
- o Vector group (connection/displacement)/Diagram
- o K-Factor rating
- o Connection Diagram
- o Impedance rating/Percentage Impedance
- o Cooling of transformer (e.g. ONAN/ONAF)
- o Protection category/type (e.g NEMA 1 Enclosure/IPxx)
- o Quantity of insulating gas/liquid (litres/gallons of oil) and type (Not applicable for Dry type cast Resin Transformer)
- o Insulation/temperature class of winding insulation.
- o Manufacturer tested date
- o Total weight (kg)

6.7.4.3 Accessories

- i. Provide fan cooling equipment (fans, power/control wiring, Local selector switch, controller, fuse, indicating light, relay) if needed.
- ii. Three-phase winding temperature indicator with contacts to control fans and alarm contacts

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- o Temperature sensing elements shall be imbedded in the windings, with means of replacement in the event of failure.
 - o In the event of an increase in transformer operating temperature, a means of alarming (sent via SCADA or locally sounded) shall occur. It is recommended the temperature alarm set point be 20 degrees C less than the overtemperature trip set point or the maximum nameplate temperature rating on units without a trip function.
 - o In the event of a temperature sensor failure, it would be preferable to fail into a safe mode (the transformer remaining in operation), with an alarming function (sent via SCADA or locally sounded) to inform the site the sensor has failed. It is further recommended any trip interlock function associated with failure of a temperature sensor be disabled, unless required by Local Code or Regulation.
- iii. Provide Class B Stranded Type SIS power/control conductors
- v. Copper ground pads
- vi. Power monitoring connection
- vii. Nameplates (Refer to 6.7.4.2)

6.7.4.4 Control power

- i. Self-powered from secondary bushing single phase
- ii. Three phase power supply, from LV distribution board, to power the cooling fans and temperature controls for the transformer. Cooling fans should be monitored, for any failure, when activated by high loading or high temperature. Any fan failure that is activated shall be monitored locally and remotely via Scada

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6.7.5 Installation

6.7.5.1 Safety:

- i. Only qualified/competent technician should work on electrical equipment.
- ii. Risk assessment should be conducted to determine the hazards and mitigations to minimize the RISK to ALARP.
- iii. Proper personal protective equipment (PPE) should be worn when performing the installation work.
- iv. Any person with pacemaker, metal implant, or wearing jewelry should not be close (3 metres/10 feet) to energized/operational transformer.
- v. Applied Electrical Safety Rules/procedures, such as safe isolation and earthing/grounding, lock-out-tag-out, permit-to-work, etc., for all work on Electrical equipment.
- vi. Take all necessary safety precautions in handling and transporting of equipment for installation such as lifting, siting, transportation route, techniques, barricade, etc., Refer to local safety requirements, owner, manufacturer and contractor safety procedures.
- vii. Ensure sound earthing/grounding of electrical equipment and metallic structure/s.

6.7.5.2 Site installation:

- i. Ventilated dry-type transformers should not be located in polluted environments and or areas with possible dripping water.
- ii. Dry-type transformers installed in high humidity location should be specified to the manufacturer and precautions taken to manage it when transformer is de-energised over a long period of time.
- iii. Location should have sufficient space, clearance to walls and no obstruction to the transformer air circulation for its cooling.
- iv. High voltage and low voltage cables connections to the transformer should be clearly marked/identifiable.

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- v. Cables should be properly clamped to avoid stress to the insulators/connectors/termination.
- vi. Recommended torque value shall be applied to all connections and bolting.
- vii. Installation of transformers and other precautions shall be in accordance to Manufacturer's instruction manual on installation and the QA/QC shall be approved by the Manufacturer's authority before energized.

6.8 Voltage Sag Mitigation System (SMS)

6.8.1 General requirement

- a) Voltage Sag Mitigation system shall have the capacity to supply the total load demand and an additional 20% of spare capacity.
- b) The SMS rating shall reflect its 100% service duty.
- c) The available fault current at the point of service will be reviewed and the SMS system and its components shall have suitable withstand rating.
- d) The SMS system and its components will meet all seismic requirements for the location where it is being installed.
- e) The level of noise produced by the SMS, operating at rated voltage and frequency, shall be below any limitations which required by local regulations.
- f) If conditions warrant, the unit will be ordered with topicalization and fungus protection.
- g) The manufacturer shall install electrical components in the SMS system in conformance with either NFPA 79 or IEC 60204.
- h) The SMS system shall be self-contained and not be dependent on outside utilities for system operation, it shall not impact the operation due to utility source power outage.
- i) The SMS system shall be designed for continuous operation for the duration designed and shall be capable of supplying start-up transient/inrush, surges, and unbalance associated with the loads it serves and when switched into service.
- j) Electrical equipment and components shall be protected from intrusion of liquids/chemicals and environment (i.e. indoor/outdoor; hazardous; dust, etc.,). Solutions include rain caps, appropriate IP or NEMA ratings, or other containment.
- k) A coordination study shall be performed to allow for selective coordination of the system protective devices.